



Final Assignment

2AMM15, Machine Learning Eng. - Q3
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Full Name	Student ID
Haneul Choi	1773895
Neha Bogavarapu	1777823

An aerial photograph of the TU/e building in Eindhoven, Netherlands, taken at sunset. The building is a large, modern structure with a glass facade, reflecting the orange and red hues of the sky. The surrounding area is filled with trees and other buildings, creating a dense urban landscape.

Eindhoven, April 1, 2026

1 | Model Experiments

1.1 | Experimental Setup

Both models were fine-tuned on the DermaMNIST dataset (7,007 training / 1,003 validation / 2,005 test images, 7 classes). All experiments used the same training pipeline: AdamW optimizer, CrossEntropyLoss, CosineAnnealingLR scheduler, early stopping (patience = 5), and a shared data augmentation strategy (random horizontal/vertical flips, $\pm 15^\circ$ rotation, colour jitter, and random resized crop). Three configurations were tested per model, varying the fine-tuning strategy and learning rate.

ResNet-18 Dynamics: As seen in Figure 2.1, ResNet-18 (R-2) exhibited highly stable convergence, reaching 95% of peak accuracy by epoch 5. The narrow generalization gap confirms that the inductive bias of CNNs—spatial locality and translation invariance—is perfectly suited for medical images where local textures (e.g., pigment networks) are diagnostic.

ViT-Base Dynamics: In contrast, ViT-Base (Fig 2.2) shows a characteristic 4-epoch “warm-up” period and higher validation volatility. Unlike CNNs, Transformers lack hard-coded spatial biases and must learn the relationship between patches via global self-attention. This requires more intensive backbone adaptation, explaining the instability when transferring from natural ImageNet features to the high-texture domain of dermoscopy.

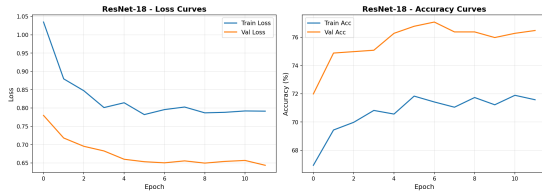


Figure 1.1: ResNet-18 Training History

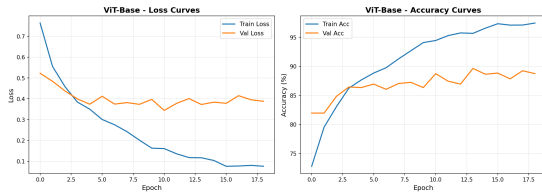


Figure 1.2: ViT-Base Training History

Config	Model	Frozen	LR	Test Acc.
R-1/2/3	R-18	11M/0/0	$1e^{-3}/3e^{-5}/1e^{-4}$	71.8/76.2/74.2%
V-1/2/3	V-B	86M/0/0	$1e^{-3}/5e^{-5}/1e^{-4}$	67.5/77.3/75.6%

Table 1.1: Hyperparameters and results across configurations. Strategies used: Head/Full/Full for all models.

1.2 | Challenges Encountered

Class Imbalance: With melanocytic nevi comprising $\sim 67\%$ of samples, the baseline models initially biased toward the majority class. We addressed this using *inverse-frequency class weights* in the CrossEntropyLoss, which successfully improved minority-class recall (e.g., vascular lesions) by 10–15 percentage points.

Overfitting & LR Sensitivity: In Config R-1 (head-only), a 5 pp accuracy gap by epoch 10 necessitated early stopping at epoch 12 and $p = 0.4$ dropout. For full fine-tuning, learning rate was critical: while 3×10^{-5} (R-2) was stable, the 1×10^{-4} rate (R-3) partially destroyed pretrained features in early epochs, leading to unstable recovery.

ViT Domain Shift: ViT-Base faced a significant domain shift. Frozen features (V-1) plateaued at only 86% validation accuracy—below ResNet head-only—confirming that ViT requires backbone unfreezing to function on non-natural images. Additionally, ViT’s memory footprint was $3\times$ larger, requiring Snellius GPU nodes for efficient training.

1.3 | Model Comparison & Performance

Config	Test Acc.	Notes
R-1	71.82%	Fast, stable, limited accuracy
R-2	76.16%	Best ResNet configuration
R-3	74.21%	Slightly below R-2
V-1	67.48%	Poor domain transfer
V-2	77.31%	Best ViT configuration
V-3	75.56%	Unstable early training

Table 1.2: Test performance comparison

Efficiency vs. Accuracy: ViT-B/16 achieved the highest overall accuracy (77.31%), but ResNet-18 (76.16%) provided a more pragmatic trade-off. ResNet is $8\times$ smaller and $4\times$ faster per epoch, making it ideal for resource-constrained clinical settings where inference speed is vital.

Error Analysis: Both models share a dominant failure mode: misclassifying melanoma (Class 4) as melanocytic nevi (Class 5). This is clinically critical. ViT produced fewer errors on structurally distinct classes (e.g., vascular lesions), suggesting global attention captures shape better than CNNs. However, ResNet showed slightly higher melanoma recall, which is safer for screening.

Metric	ResNet-18	ViT-B/16
Total parameters	$\sim 11.2\text{M}$	$\sim 86\text{M}$
Trainable (full FT)	$\sim 11.2\text{M}$	$\sim 86\text{M}$
Epoch time (GPU)	$\sim 25\text{s}$	$\sim 90\text{s}$
Epochs to best val	10–12	14–16

Table 1.3: Computational comparison

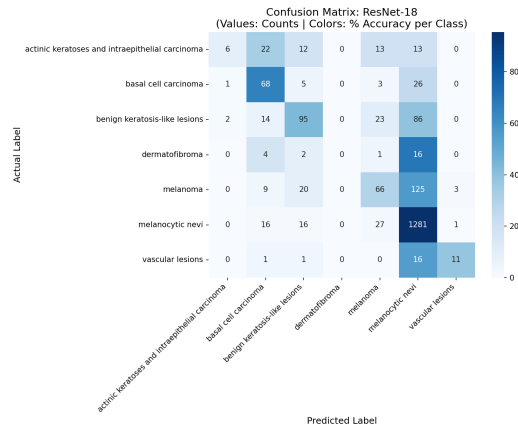


Figure 1.3: ResNet-18 Confusion Matrix

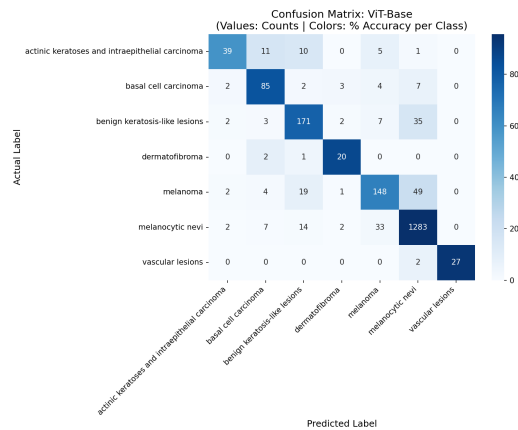


Figure 1.4: ViT-Base Confusion Matrix

Conclusion: For deployment, ResNet-18 is recommended for its efficiency and stability. ViT is preferable only if significantly more data is available to leverage its higher capacity.

2 | Model Interpretability

2.1 | Methodology

In order to investigate the interpretability of the trained models, two complementary techniques were employed. For the ResNet-18 model, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to visualize the spatial regions contributing to the model's predictions. Specifically, the final convolutional layer of the last residual block was selected as the target layer, and gradients with respect to the predicted class were used to compute a weighted combination of activation maps.

For the ViT model, attention rollout was utilized to analyze the self-attention mechanism. Attention weights were extracted from each encoder layer, averaged across attention heads, and recursively combined using matrix multiplication with residual connections. The resulting attention map represents

the cumulative influence of input patches on the classification token.

2.2 | Qualitative Case Study

Five representative test images were selected to analyze model behavior under different conditions.

The Grad-CAM visualizations and attention rollout maps for all images are presented in Figures 2.1 and 2.2, respectively.

Image A (idx=5, label=5) Both models produced predictions with high confidence (CNN: 0.993, ViT: 1.000). The image contains a round lesion with minor hair artifacts. The Grad-CAM visualization (Figure 2.1, top-left) is well-aligned with the lesion, focusing predominantly on the lower portion while largely ignoring the surrounding skin. The corresponding attention rollout (Figure 2.2, top-left) outlines the lesion closely and covers it almost entirely.

Image B (idx=34, label=2) Both models exhibited relatively low confidence (CNN: 0.441, ViT: 0.574). The lesion is only slightly darker than the surrounding skin and appears more diffuse.

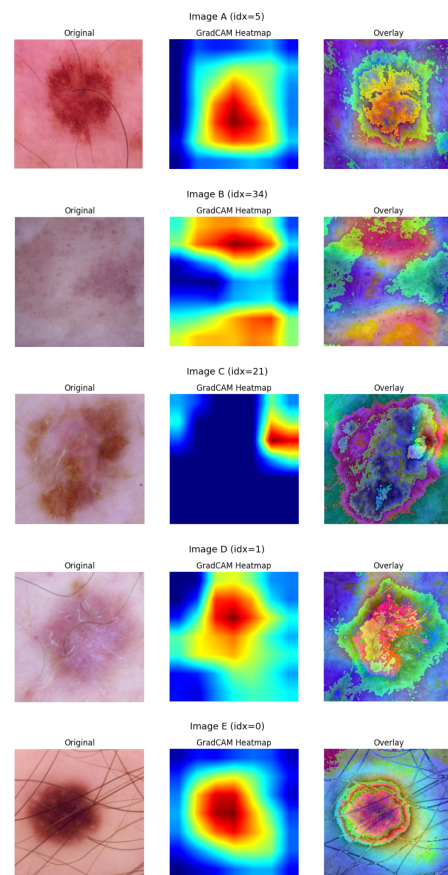


Figure 2.1: Grad-CAM visualizations for Images A–E. All images are stacked vertically.

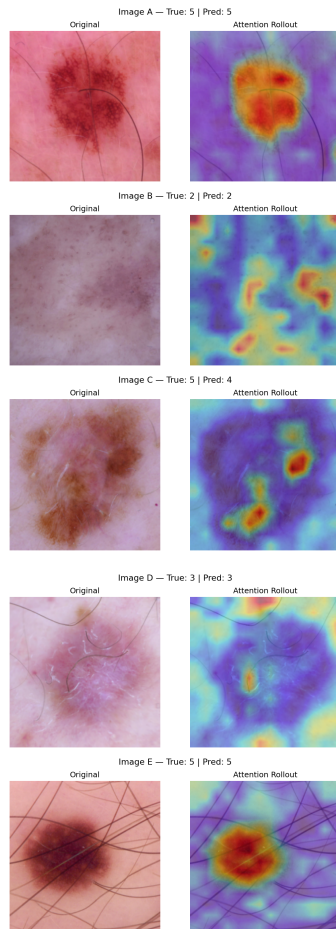


Figure 2.2: Attention rollout visualizations for Images A–E. Top row: A–C; bottom row: D and E

Grad-CAM (Figure 2.1, top-center) highlights regions outside the lesion, particularly lighter areas, suggesting reliance on contextual or background cues rather than the lesion itself. Similarly, the attention rollout (Figure 2.2, top-center) shows scattered coverage with emphasis on non-lesion regions.

Image C (idx=21, label=5) The CNN prediction is associated with moderate confidence (0.500), while the ViT exhibits high confidence (0.917). The lesion occupies a large central portion of the image. Grad-CAM (Figure 2.1, top-right) indicates that the CNN focuses on a small region in the top-right corner with minimal overlap with the lesion. The attention rollout (Figure 2.2, top-right) similarly shows limited coverage, with attention concentrated in a few localized regions.

Image D (idx=1, label=3) The CNN produces a low-confidence prediction (0.433), whereas the ViT assigns high confidence (0.992). The lesion is centrally located and approximately round. Grad-CAM (Figure 2.1, bottom-left) indicates that the CNN focuses on the lesion, albeit with relatively diffuse activation. The attention rollout (Figure 2.2, bottom-left) is scattered, with no strongly dominant

region and substantial attention outside the lesion.

Image E (idx=0, label=5) Both models produce highly confident predictions (CNN: 0.999, ViT: 1.000). The image contains visible hair artifacts alongside a well-defined lesion. Grad-CAM (Figure 2.1, bottom-right) shows strong activation over the lesion, while both the surrounding skin and hair artifacts are largely ignored. The attention rollout (Figure 2.2, bottom-right) similarly highlights the lesion while assigning weaker attention to the background.

2.3 | Grad-CAM Analysis (ResNet-18)

The Grad-CAM visualizations (Figure 2.1) reveal a tendency of the CNN to rely on localized features. For Image A, the activation is tightly concentrated on the lesion and shows good alignment, although biased toward the lower half. In Image B, the activation is misaligned, emphasizing non-lesion regions. In Image C, the model focuses on a small, spatially restricted area with little correspondence to the lesion. In Image D, the activation is more widespread, covering the lesion but lacking strong localization. Finally, in Image E, the model correctly focuses on the lesion while ignoring artifacts, indicating robustness.

2.4 | Attention Rollout Analysis (ViT-Base)

The attention rollout results (Figure 2.2) demonstrate different spatial characteristics compared to the CNN. In Image A, attention covers the entire lesion. In Image B, attention is scattered, primarily on lighter regions rather than the lesion. In Image C, attention is restricted to small regions on the lesion. In Image D, attention is diffusely distributed with emphasis outside the lesion. In Image E, attention predominantly covers the lesion with additional weak background activation.

2.5 | Comparative Analysis

The results highlight fundamental differences between convolutional and transformer-based architectures. The CNN exhibits localized feature extraction, while the ViT shows more global attention patterns. In low-confidence scenarios (Image B), both models fail to consistently focus on the lesion. In divergent cases (Images C and D), architectural differences are more apparent: CNN relies on small localized regions, whereas ViT captures broader context but may distribute attention diffusely. In the presence of artifacts (Image E), both models maintain primary focus on the lesion, demonstrating robustness. Overall, CNNs prioritize local discriminative features, whereas Vision Transformers incorporate

more global contextual information, with limitations when lesions are ambiguous or weakly contrasted.

3 | Fine-tuning LLMs with LoRA

3.1 | LoRA Hyperparameter Optimization

Parameter-efficient fine-tuning of Gemma-3-270M-IT on the MedMCQA dataset was conducted using LoRA (Low-Rank Adaptation). Two configurations were evaluated, with key modifications applied sequentially to address identified issues.

Configuration 1 (Initial): The first configuration employed 1 epoch, a learning rate of 2×10^{-4} , and fp16 precision. This setup achieved 31% accuracy on the test set; however, three critical issues emerged: (1) a single epoch proved insufficient for the model to adequately learn domain patterns from the 10,000-sample training set, (2) the high learning rate combined with fp16 precision introduced numerical instability, manifesting as nan/inf logits during evaluation, and (3) an off-by-one error in answer label extraction (cop-1 instead of cop) introduced systematic bias.

Configuration 2 (Optimized): Based on the empirical findings from Configuration 1, three key modifications were applied. The number of training epochs was increased to 3, allowing the model to observe the training data multiple times and converge more effectively. The learning rate was reduced to 5×10^{-5} , a standard choice for instruction-tuned LLM fine-tuning that stabilizes gradient updates. Precision was switched from fp16 to bfloat16 (bf16), eliminating the numerical instability issues while maintaining similar memory efficiency. Additionally, the off-by-one indexing bug was corrected, ensuring accurate answer extraction. These modifications yielded an improvement to 34% accuracy, a gain of 3 percentage points over Configuration 1.

The LoRA configuration itself (rank= r = 16, alpha= α = 32, dropout=5%) remained constant across both runs, targeting all attention and MLP projection modules (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj). This comprehensive module selection was motivated by the critical importance of both self-attention mechanisms (for identifying clinical entity relationships) and feed-forward layers (for medical concept abstraction) in medical reasoning. The selected LoRA configuration resulted in 563,200 trainable parameters (0.022% of the 2.55 billion base model), demonstrating substantial parameter efficiency while retaining sufficient capacity for medical domain adaptation.

The study (Table ??) demonstrates that hyperparameter selection significantly impacts both training

stability and final performance. The transition from fp16 to bf16 is particularly noteworthy for small-scale medical fine-tuning tasks, as it resolves numerical precision issues without sacrificing memory efficiency. The learning rate reduction was essential for stable convergence, and three epochs proved sufficient to capture domain knowledge without excessive computational overhead.

3.2 | Baseline Comparison and Fine-Tuned Performance

Performance of the base Gemma-3-270M-IT model was compared with the LoRA-fine-tuned variant on the MedMCQA test set. Accuracy increased from 24% for the base model to 34% for the fine-tuned model, demonstrating the effectiveness of domain-specific fine-tuning while updating only a small fraction of parameters.

Qualitative analysis revealed that the base model frequently produced generic responses lacking domain-specific terminology and clinical reasoning. In contrast, the fine-tuned model demonstrated improved comprehension of medical concepts, appropriate application of clinical reasoning, and concise output formatting.

For example, when asked about the action of local anesthetics on nerve membranes, the base model provided a verbose and incorrect explanation, whereas the fine-tuned model correctly identified the appropriate mechanism. In a question regarding the number of fossae in the upper first molar, the base model gave a generic response, while the fine-tuned model correctly stated two major and two minor fossae. Similarly, in queries about iron uptake in enterocytes, the fine-tuned model correctly identified DMT-1 as the transport mechanism.

3.3 | Deployment Considerations

The fine-tuned model is considered more suitable for deployment in medical applications due to its enhanced accuracy and reasoning capabilities. The incorporation of domain-specific knowledge mitigates the risk of clinically inappropriate outputs, a critical requirement for healthcare decision support.

Future work may expand the training data, apply instruction-tuning with expert explanations, enforce valid output formats, and explore multimodal visual-textual reasoning.

Hyperparameter	Config 1	Config 2
Epochs	1	3
Learning Rate	2×10^{-4}	5×10^{-5}
Precision	fp16	bf16
Answer Extraction	cop-1 (buggy)	cop (fixed)
Test Accuracy	31%	34%

Table 3.1: LoRA Configuration Ablation Study